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Linear rank tests with right censored data

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Summary

Linear rank statistics are developed for tests on regression coefficients with censored data. These statistics arise as score statistics based on the marginal probability of a generalized rank vector. The observed Fisher information provides a variance estimator generally, while in certain special cases a permutation approach to variance estimation is also possible. The Wilcoxon test is generalized to regression tests with censored data. The log rank test (Cox, 1972; Peto & Peto, 1972) is also produced as a special case.

Some key words: Approximate score; Censored data; Linear regression; Log rank test; Rank test; Wilcoxon test.

1. Introduction

Consider a linear regression model

$$y = \alpha + \beta z + \sigma e, \tag{1}$$

with error probability density function $f(e)$. The regression variable z has been written as a column p vector with corresponding row-vector, β, of coefficients. Suppose that the response variable is subject to right censoring via a mechanism that is independent of that giving rise to y. A test of $\beta = \beta_0$ is frequently of interest. For example, the hypothesis of equality of $p+1$ samples can be formulated as the hypothesis $\beta = 0$, with z given by indicator variables corresponding to any p of the samples.

An assumed f allows parametric likelihood results to be derived, leading to a score test of $\beta = \beta_0$. For comparison with the rank tests that follow, this test will be written in terms of ordered residuals. Let $w = y - \beta_0 z$ represent a residual about the hypothesized value. Let $w_{(1)} < \ldots < w_{(k)}$ represent the ordered uncensored residuals in the sample. In some arbitrary order, let $w_{(i1)}, \ldots, w_{(im_i)}$ denote the right censored residuals in $[w_{(i)}, w_{(i+1)})$ for $i = 0, 1, \ldots, k$, with $w_{(0)} = -\infty$, $w_{(k+1)} = \infty$. Suppose, as usual, that the underlying unobserved residuals are known only to exceed the corresponding censoring points. Also let $z_{(i)}$ and $z_{(ij)}$ represent regression vectors corresponding to $w_{(i)}$ and $w_{(ij)}$ respectively. A comma will be used to separate the subscripts of w and of z if the possibility of ambiguity exists. Note that uncensored residuals are assumed not to tie with each other, while right censored residuals may tie with each other or with uncensored residuals. To fix the somewhat nonstandard notation, consider the following simple numerical example. Suppose that the calculated values of w are $(112, 69^+, 32, 112^+)$ with $^+$ indicating censoring, and that the corresponding scalar regressor values are, respectively, $(1{\cdot}4, 0{\cdot}0, 0{\cdot}7, 1{\cdot}4)$. We then have $w_{(1)} = 32$, $w_{(2)} = 112$, $w_{(1,1)} = 69$, $w_{(2,1)} = 112$; $z_{(1)} = 0{\cdot}7$, $z_{(2)} = 1{\cdot}4$, $z_{(1,1)} = 0{\cdot}0$, $z_{(2,1)} = 1{\cdot}4$.

The log likelihood, up to an additive constant, may be written

$$l = \sum_{i=0}^{k} \left(\log f[\{w_{(i)} - \alpha - (\beta - \beta_0) z_{(i)}\} \sigma^{-1}] + \sum_{j=1}^{m_i} \log F[\{w_{(ij)} - \alpha - (\beta - \beta_0) z_{(ij)}\} \sigma^{-1}] \right) - k \log \sigma, \tag{2}$$

where, again in somewhat nonstandard notation, $F(v)$ is used for one minus the distribution

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function rather than for the distribution function itself; that is

$$F(v) = \int_v^\infty f(u)\,du.$$

Also, by convention, $\log f(-\infty) = 0$ and any summation $a_1 + \ldots + a_m$ has value zero if $m = 0$. Expression (2) can be rewritten

$$l = \sum_{i=0}^{k} \left\{ \log f(\tau_{(i)} - \gamma z_{(i)}) + \sum_{j=1}^{m_i} \log F(\tau_{(ij)} - \gamma z_{(ij)}) \right\} - k \log \sigma,$$

where $\tau_{(i)} = (w_{(i)} - \alpha)\,\sigma^{-1}$, $\tau_{(ij)} = (w_{(ij)} - \alpha)\,\sigma^{-1}$ and $\gamma = (\beta - \beta_0)\,\sigma^{-1}$. A score test for $\beta = \beta_0$, or equivalently $\gamma = 0$, utilizes the statistic

$$[dl/d\gamma]_{\gamma=0} = \sum_{i=0}^{k} \left(z_{(i)} c_i + \sum_{j=1}^{m_i} z_{(ij)} C_{ij} \right), \tag{3}$$

where $c_0 = 0$ and

$$c_i = -d \log f(\tau_{(i)})/d\tau_{(i)}, \quad C_{ij} = -d \log F(\tau_{(ij)})/d\tau_{(ij)} \tag{4}$$

are scores corresponding respectively to uncensored and censored residuals. The scores (4) depend, through values of τ, on α and σ, which must be estimated before the test statistic (3) can be calculated. Usually maximum likelihood estimators of α and σ, subject to $\beta = \beta_0$, are used, in which case the statistic (3) is asymptotically fully efficient under the assumed f.

A usual uncertainty concerning the choice of f, and the possibility that a few outlying residuals may have a dominating effect on (3) are important reasons for seeking alternatives to (3). With uncensored data, rank tests generally possess greater robustness against a wrong choice of f and greater outlier resistance than do the corresponding parametric tests (3). In § 2 tests based on a censored data generalization of the ranks of w are derived and are shown to be of precisely the same form as (3), with c_i and C_{ij} obtained from a score generating distribution. One attractive feature of the rank invariant tests is that the scores corresponding to (4) are independent of α and σ, so that the problem of estimating α and σ does not arise. Because the construction of the censored data test statistic is essentially the same as that for uncensored data, it is appropriate to first review the construction and properties of locally optimal rank tests with uncensored data.

Suppose, in the above notation, that there is no possibility of censoring, so that all $m_i = 0$, and the total sample size is $n = k$. The rank vector $r = r(w)$ is given by the labels corresponding to the ordered residuals; that is, $r(w) = \{(1), \ldots, (n)\}$. The rank vector probability is precisely

$$p(r) = \int \prod_{i=1}^{n} f(\tau_{(i)} - \gamma z_{(i)})\,d\tau_{(i)}, \tag{5}$$

where here and later the integral is over the region $\tau_{(1)} < \ldots < \tau_{(n)}$. A locally most powerful rank test can be based on the score statistic for $\gamma = 0$, that is $\beta = \beta_0$, derived from (5) (Puri & Sen, 1971, p. 108; Cox & Hinkley, 1974, p. 188). Straightforward calculations give

$$v = [d \log p(r)/d\gamma]_{\gamma=0} = \sum_{i=1}^{n} z_{(i)} c_i, \tag{6}$$

where

$$\begin{aligned} c_i &= n! \int \{-d \log f(\tau_{(i)})/d\tau_{(i)}\} \prod_{j=1}^{n} \{f(\tau_{(j)})\,d\tau_{(j)}\} \\ &= n! \int \phi(u_i) \prod_{j=1}^{n} du_j \\ &= E\{\phi(u_i)\}. \end{aligned} \tag{7}$$

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Note that $u_i = 1-F(\tau_{(i)})$ is the ith order statistic in a uniform $(0,1)$ sample of size n, $\phi(u)$ for $0<u<1$ is given by

$$\phi(u) = \phi(u,f) = -f'\{F^{-1}(1-u)\}/f\{F^{-1}(1-u)\}.$$

Note that the sum of scores and the expectation of v are zero, because the sum over all ordered values can be replaced by a sum over the unordered values. The fact that u_i has expectation $i(n+1)^{-1}$ for $i = 1,\ldots,n$ leads to an asymptotically equivalent system of scores given by

$$c_i = \phi\{i(n+1)^{-1}\}. \tag{8}$$

Some interesting special cases of (7) and (8) are as follows. A logistic density,

$$f(\tau) = e^{\tau}(1+e^{\tau})^{-2},$$

gives Wilcoxon (1945) scores for both (7) and (8), $c_i = 2i(n+1)^{-1}-1$. A standard normal density gives the normal scores test (7), $c_i = E(\tau_{(i)})$ (Fisher & Yates, 1963) and the van der Waerden (1953) test (8), $c_i = G^{-1}\{i(n+1)^{-1}\}$, where G is the standard normal distribution function. Similarly, an extreme minimum value density, $f(\tau) = \exp(\tau-e^{\tau})$, yields, for (7), the exponential scores $c_i = n^{-1}+(n-1)^{-1}+\ldots+(n-i+1)^{-1}-1$ of Savage (1956), while the double exponential density, $f(\tau) = \frac{1}{2}e^{-|\tau|}$ gives sign, i.e. median, test scores (8), $c_i = \operatorname{sgn}\{2i-(n+1)\}$, with $c_i = 0$ if $i = \frac{1}{2}(n+1)$.

The significance level based on (6) can be assessed in terms of the permutation distribution of v or approximately by noting that the variance matrix, V, is

$$(n-1)^{-1}\sum_{i=1}^{n} c_i^2(Z^{\mathrm{T}}Z),$$

where Z is the $n\times p$ matrix of regression vectors centred about their means. More specifically, under mild restrictions on the regression vectors (Hájek & Šidák, 1967, p. 159), v is asymptotically normal, so that $v^{\mathrm{T}}V^{-1}v$ is asymptotically χ_p^2, under $\beta = \beta_0$, assuming that Z is of rank p.

Like the parametric test (3), the rank test (6) is asymptotically fully efficient if the assumed score generating distribution, f, agrees, up to location and scaling, with the actual sampling distribution, say f_0. More generally the asymptotic relative efficiency (Hájek & Šidák, 1967, p. 268) is the square of the limiting correlation of (7) or (8) with the locally optimum test based on f_0. The efficiency may be written

$$\left\{\int_0^1 \phi(u,f)\,\phi(u,f_0)\,du\right\}^2 \Bigg/ \left\{\int_0^1 \phi^2(u,f)\,du \int_0^1 \phi^2(u,f_0)\,du\right\}, \tag{9}$$

assuming the Fisher information terms in the denominator are both finite. Under $f\neq f_0$, (9) typically indicates important improvements over the corresponding parametric test. For example, the normal scores test, under mild conditions on f_0 (Puri & Sen, 1971, p. 118), has efficiency equal or greater than that of the corresponding least squares test. Under Cauchy sampling, for instance, the least squares procedure has efficiency zero, while the normal scores test has efficiency 0·43. Rank tests themselves differ somewhat in efficiency properties. For example, the Wilcoxon test has efficiency 0·61 under Cauchy sampling while the sign test has even higher efficiency 0·81. It is then important to consider carefully plausible alternatives to the score generating distribution in selecting a rank test. Birnbaum & Laska (1967) and Gastwirth (1970) give some consideration to the selection of robust rank tests.

There are several possibilities for extending the definition of the rank vector to right censored data. Perhaps the most comprehensive definition (Peto, 1972a) takes the rank vector to consist of the labels corresponding to the ordered sample of all censored and

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uncensored values of w along with the associated censoring indicators. This statistic is the maximal invariant statistic under monotone increasing transformations on w. It, however, has a distribution that depends in a complicated manner on the censoring mechanism (Crowley, 1974). An alternative (Kalbfleisch & Prentice, 1973) takes the point of view that it is the rank vector of the underlying uncensored values of w, that are only partially observed because of the censoring, that is of primary interest. For instance, the rank vector underlying the simple numerical example above is known to be an element of

$$\{(3,2,1,4),\ (3,1,2,4),\ (3,1,4,2)\}$$

according to whether the value of w censored at 69 is the third largest, second largest or largest residual. The rank vector probability may then be taken to be the sum of the probabilities of the possible underlying rank vectors, given the data. This definition, unlike that given above, does not utilize the ordering of censored observations between adjacent uncensored values. It can be specified by the labels $(1), \ldots, (k)$ of ordered uncensored values of w as well as the labels $\{(i1), \ldots, (im_i)\}$ of censored values in $[w_{(i)}, w_{(i+1)})$ in some arbitrary order for $i = 0, \ldots, k$. The rank vector, generalized to censored data, will therefore be denoted

$$r = r(w) = [(1), \ldots, (k);\ \{(i1), \ldots, (im_i)\}, i = 0, \ldots, k]. \tag{10}$$

In § 2, the probability, $P(r)$, of possible underlying rank vectors consistent with (10), will be used to derive linear rank statistics of the form (3). An approximate scoring system, generalizing (8), is also given. A variance estimator, appropriate with censoring that may depend on z, is given in § 3 followed by discussion and illustration.

2. Linear rank statistics with censored data

2·1. *Test construction*

By the method of Kalbfleisch & Prentice (1973), the probability that the underlying rank vector is one of those possible, given the sample, can be calculated in two steps: with the notation of § 1 the accumulated probability of possible rankings for censored residuals, conditional on the uncensored residuals, $w_{(1)} < \ldots < w_{(k)}$, is simply

$$\prod_{i=0}^{k} \prod_{j=1}^{m_i} F(\tau_{(i)} - \gamma z_{(ij)}). \tag{11}$$

This expression arises because the possible rankings corresponding to each censored residual $w_{(ij)}$ map out an integral over the whole line segment $[w_{(i)}, \infty)$. Note that the term with $i = 0$ can be dropped as $\tau_{(0)} = -\infty$. The unconditional probability, $P(r)$, that is the accumulated probability of possible underlying rank vectors, is then

$$P(r) = \int \prod_{i=1}^{k} \left\{ f(\tau_{(i)} - \gamma z_{(i)}) \prod_{j=1}^{m_i} F(\tau_{(i)} - \gamma z_{(ij)})\, d\tau_{(i)} \right\}. \tag{12}$$

Because the τ's enter (12) as dummy variables, $P(r)$ is completely independent of α and σ. Note that at $\gamma = 0$, or $\beta = \beta_0$, (12) can be integrated directly giving that at $\gamma = 0$, the value of $P(r)$ is $(n_1 \ldots n_k)^{-1}$, where $n_i = (m_i + 1) + \ldots + (m_k + 1)$ is the number of individuals with values of w known to be equal to or greater than $w_{(i)}$. Also, the number of underlying rank vectors consistent with r is precisely $n!/(n_1! \ldots n_k!)$.

As in (6), a score statistic for testing $\beta = \beta_0$ may be based on (12); Johnson & Mehrotra (1972) give a simple special case. Thus

$$v = [d \log P(r)/d\gamma]_{\gamma=0} = \sum_{i=1}^{k} (z_{(i)} c_i + s_{(i)} C_i), \tag{13}$$

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where $s_{(i)} = z_{(i1)} + \ldots + z_{(im_i)}$, c_i is a score corresponding to $w_{(i)}$ and C_i is a score corresponding to each of $w_{(i1)}, \ldots, w_{(im_i)}$. Assuming sufficient regularity that differentiation and integration may be interchanged, the censored and uncensored data scores are respectively

$$c_i = \int \{-d\log f(\tau_{(i)})/d\tau_{(i)}\} \prod_{j=1}^{k} \{n_j F^{m_j}(\tau_{(j)}) f(\tau_{(j)})\, d\tau_{(j)}\}, \tag{14}$$

$$C_i = \int \{-d\log F(\tau_{(i)})/d\tau_{(i)}\} \prod_{j=1}^{k} \{n_j F^{m_j}(\tau_{(j)}) f(\tau_{(j)})\, d\tau_{(j)}\}, \tag{15}$$

where both integrals are again over the region $\tau_{(1)} < \ldots < \tau_{(k)}$. A simplified form arises upon setting $u_j = 1 - F(\tau_{(j)})$ for $j = 1, \ldots, k$ and defining for $0 < u < 1$

$$\phi(u) = -f'\{F^{-1}(1-u)\}/f\{F^{-1}(1-u)\}, \quad \Phi(u) = (1-u)^{-1} f\{F^{-1}(1-u)\}, \tag{16}$$

whereupon

$$\begin{aligned} c_i &= \int \phi(u_i) \prod_{j=1}^{k} \{n_j(1-u_j)^m\, du_j\}, \\ C_i &= \int \Phi(u_i) \prod_{j=1}^{k} \{n_j(1-u_j)^{m_j}\, du_j\}, \end{aligned} \tag{17}$$

where both integrals are over the region $u_1 < \ldots < u_k$.

The statistic (13) is of the same form as the corresponding parametric test (3), though, unlike (3), the same score is assigned to all censored residuals between adjacent uncensored values.

2·2. *Special scoring systems*

Let

$$J\{g(u_i)\} = \int g(u_i) \prod_{j=1}^{k} \{n_j(1-u_j)^{m_j}\, du_j\}$$

for an arbitrary function g. A simple calculation then gives

$$J\{(1-u_i)^l\} = \prod_{j=1}^{i} \left(\frac{n_j}{n_j+l}\right) \quad (l = 1, 2, \ldots). \tag{18}$$

Also let $F_W(w) = F\{(w-\alpha)\sigma^{-1}\}$ represent the survivor function at w, given $\beta = \beta_0$. Then

$$\hat{F}_W(w) = \Pi_j^*\{n_j/(n_j+1)\}, \tag{19}$$

where the product is for $w_{(j)} \leqslant w$, will, under mild conditions on the censoring, provide a consistent estimator of F_W, under $\beta = \beta_0$, as can be seen by comparison with the product-limit estimator (Kaplan & Meier, 1958)

$$\Pi_j^*\{(n_j-1)/n_j\} \tag{20}$$

or Altshuler's (1970) estimator

$$\tilde{F}_W(w) = \exp(-\Sigma_j^* n_j^{-1}). \tag{21}$$

Now return to the scores (17). A logistic density gives $\phi(u) = 2u-1$, $\Phi(u) = u$, so that, from (18) and (19),

$$c_i = 1 - 2\hat{F}_W(w_{(i)}), \quad C_i = 1 - \hat{F}_W(w_{(i)}). \tag{22}$$

Peto & Peto (1972) gave a generalization of the Wilcoxon and Kruskal–Wallis tests similar to those arising from (22). In comparison the scores utilized by Gehan (1965) and Breslow (1970) for a censored data generalization of the Wilcoxon test can be written $c_i = i - n_i$, $C_i = i$. Peto & Peto indicate that the Gehan scores will give rise to poorer local power properties at the logistic model than will (22).

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Similarly, an extreme minimum value density, $f(\tau) = \exp(\tau - e^{\tau})$, yields

$$\phi(u) = -\log(1-u) - 1, \quad \Phi(u) = -\log(1-u).$$

Direct integration gives $J\{\log(1-u)\} = \log \tilde{F}_W(w_{(i)})$, so that

$$c_i = -\log \tilde{F}_W(w_{(i)}) - 1, \quad C_i = -\log \tilde{F}_W(w_{(i)}). \tag{23}$$

These so-called log rank scores were given by Mantel (1966) and Peto & Peto (1972) for the comparison of two or several samples and by Cox (1972) for a global test of no regression within the proportional hazards model. J. Crowley and D. R. Thomas in a University of Wisconsin technical report found the scores (22) and (23) for the K-sample problem by an approximation to the distribution of the rank vector for combined censored and uncensored observations conditional on the vector of censoring indicators. Note that the statistic (13) with scores (23) can be written in a notation similar to that used by Cox (1972) as

$$v = \sum_{i=1}^{k} n_i^{-1} \sum_{l \in R(w_{(i)})} z_l - \sum_{i=1}^{k} z_{(i)},$$

where $R(w_{(i)})$ is the set of individuals with values of w equal to or greater than $w_{(i)}$.

The formulae (17) are inconvenient for many other score generating densities f. A procedure for approximate scoring is indicated in § 3·4.

3. Properties of the score statistic v

3·1. *Approach*

With progressive type II censoring, specified numbers of individuals are censored at the ordered residuals $w_{(1)}, \ldots, w_{(k)}$. In this case $P(r)$ is precisely the probability of observing the generalized rank vector r. It follows from usual parametric likelihood considerations that v will have mean zero and that the observed Fisher information $V_0 = -d^2 \log P(r)/d\gamma^2$ will generally give a consistent estimator of variance at $\beta = \beta_0$. Also, these properties hold regardless of the actual sampling density and regardless of whether or not the procedure for selective censoring depends on z.

This section, therefore, concentrates on the more usual type I or 'time' censoring in which each underlying w value, w_i, has a corresponding potential censoring value w_i^*. Then $P(r)$ is the probability of the set of underlying values of w that give rank vectors consistent with the observed generalized rank vector r. With 'time' censoring, however, r will be observed only in a subset of this space. Other w values in this set will give rise to a different set of censored residuals and correspondingly a different generalized rank vector. Consequently special justification is required to demonstrate that the statistic (13) has mean vector 0 and variance estimator V_0. This justification is supplied by relating the mean and variance of the censored test statistic (13) to the corresponding uncensored test statistic (6), generated by the same density function f.

3·2. *Expectation of v*

Consider a test statistic, v, of the form (13) that generated as the average of possible values of an uncensored rank statistic (6) with scores $c_1, \ldots, c_n$, $\Sigma c_i = 0$. By induction, we show that $E(v) = 0$, at $\beta = \beta_0$, or, more precisely, that $E(v \mid B_i) = 0$ for $i = 1, 2$. The condition B_i means that the underlying uncensored w values are in B_i, where $B_1 = \{(w_1, \ldots, w_n) \mid \text{all } w_i > w^*\}$, $B_2 = B_1^c$ and $w^* = \min(w_1^*, \ldots, w_n^*)$. A simple calculation shows that $E(v \mid B_i) = 0$ for samples of size two. Suppose that $E(v \mid B_i) = 0$ for $i = 1, 2$, for all censored data rank statistics, generated as described above, from samples of size less than n. Consider a sample of size n.

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Within B_1 at least one censored value, w^*, precedes any uncensored value. Such censored values are assigned a score of zero. Write $v = 0 + v_1$, where v_1 is a censored data test statistic of the form (13) based on a sample of size $n-1$ and generated by an uncensored rank test with scores $c'_i = \{(n-i)\,c_i + ic_{i+1}\}\,n^{-1}$, for $i = 1, \ldots, n-1$. Because the induction hypothesis applies to v_1, $E(v \mid B_1) = 0$. Similarly, in B_2 an uncensored value $w_{(1)}$ on study subject (1) precedes any censored w value. One can write $v = z_{(1)}c_1 - (n\bar{z} - z_{(1)})\,c_1(n-1)^{-1} + v_2$, where, given the value of (1), v_2 is a censored data test statistic on a sample of size $n-1$ generated by an uncensored test statistic with scores $c'_i = c_{i+1} + c_1(n-1)^{-1}$ for $i = 1, \ldots, n-1$. The induction hypothesis gives

$$E\{v \mid B_2, (i)\} = z_{(1)}\,c_1 - (n\bar{z} - z_{(1)})\,c_1(n-1)^{-1}. \tag{24}$$

Since each of the n study subjects has an equal probability under $\beta = \beta_0$ of giving rise to the first uncensored w value, (24) may be averaged to give $E(v \mid B_2) = 0$. It follows that $E(v) = 0$ as desired.

Note also that the sum of scores given by any realization of (10) is identically zero; that is $\Sigma\,(c_i + m_i C_i) = 0$.

This sum is simply the average of sums of scores corresponding to possible underlying rank vectors, each of which from § 1 has value zero. It is interesting to observe further that even though the unconditional expectation of v is identically zero the conditional expectation of v given the specific scores (17), or equivalently, given $\{m_0, \ldots, m_k\}$ is in general not zero. If censoring is independent of z, however, a permutation argument yields zero for the conditional expectation as well.

3·3. *Variance estimation*

A permutation approach is commonly used to calculate the variance of a linear rank statistic. If censoring does not depend on z, however, a permutation argument yields for (13) a variance matrix of

$$(n-1)^{-1} \sum_{i=1}^{k} (c_i^2 + m_i C_i^2)\, Z^{\mathrm{T}} Z,$$

though the reference set consists only of experiments generating the same set of scores.

More generally, however, a permutational argument is inappropriate. In view of the form of (12) it is natural to consider the observed Fisher information based on (12),

$$V_0 = -d^2 \log P(r)/d\gamma^2, \tag{25}$$

at $\gamma = 0$, for variance estimation. Once again special justification will be required with type I censoring.

Consider the uncensored rank probability $p(r)$ of (5). It is straightforward to note that the expectation of $-d^2 \log p(r)/d\gamma^2$ equals the variance of (6) where, as in the remainder of this subsection, evaluation at $\gamma = 0$ will be suppressed. On the other hand, straightforward differentiation of (5) yields

$$-d^2 \log p(r)/d\gamma^2 = vv^{\mathrm{T}} - n!\, d^2 p(r)/d\gamma^2.$$

Since the expectation of vv^{T} is the variance of v it follows that the expectation of $d^2 p(r)/d\gamma^2$ is zero.

Now return to the censored data generalized rank probability $P(r)$ of (12). Expression (25) can be rewritten

$$V_0 = vv^{\mathrm{T}} - (n_1 \ldots n_k)\, d^2 P(r)/d\gamma^2.$$

The expectation of vv^{T} is the variance of (13) so that it is only necessary to show that the expectation of the second term is the zero matrix. For this, note that $(n_1 \ldots n_k)\, d^2 P(r)/d\gamma^2$ is

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precisely the average of $n!\,d^2p(r)/d\gamma^2$ matrices corresponding to possible underlying uncensored rank vectors. Such matrices can be written $\Sigma\Sigma z_{(i)}\,z_{(j)}{}^{\mathrm{T}}\,a_{ij}$, where $a_{ij} = a_{ji}$, $\Sigma a_{ii} = 0$, $\Sigma\Sigma a_{ij} = 0$ and the summations are over $i, j = 1, \ldots, n$. An inductive argument very similar to that in § 3·2, and using the result of § 3·2, gives $E\{(n_1 \ldots n_k)\,d^2P(r)/d\gamma^2 \mid B_i\} = 0$. It follows that the expectation of V_0 is the variance of (13).

Consider now the calculation of V_0. After straightforward differentiation of (12), (25) can be written

$$V_0 = \sum_{i=1}^{k} [z_{(i)}\,z_{(i)}^{\mathrm{T}}\,J\{\psi_1(u_i)\} + \sum_{j=1}^{m_i} z_{(ij)}\,z_{(ij)}^{\mathrm{T}}\,J\{\psi_2(u_i)\}] - \{J(bb^{\mathrm{T}}) - vv^{\mathrm{T}}\}, \tag{26}$$

where

$$\psi_1(u) = [-d^2 \log f(\tau)/d\tau^2]_{\tau = F^{-1}(1-u)}, \quad \psi_2(u) = [-d^2 \log F(\tau)/d\tau^2]_{\tau = F^{-1}(1-u)},$$

$$b = \sum_{i=1}^{k} \{z_{(i)}\,\phi(u_i) + s_{(i)}\,\Phi(u_i)\}. \tag{27}$$

The matrix V_0 can be calculated explicitly in the special cases listed above. A logistic generating density f gives the scores (22) with variance estimator

$$V_0 = \sum_{i=1}^{k} \left\{ a_i(1-a_i^*) \left(2z_{(i)}\,z_{(i)}^{\mathrm{T}} + \sum_{j=1}^{m_i} z_{(ij)}\,z_{(ij)}^{\mathrm{T}} \right) - (a_i^* - a_i)\,x_{(i)} \left(a_i\,x_{(i)}^{\mathrm{T}} + 2 \sum_{j=i+1}^{k} a_j\,x_{(j)}^{\mathrm{T}} \right) \right\}, \tag{28}$$

where

$$a_i = \hat{F}_W(w_{(i)}) = \prod_{j=1}^{i} \left(\frac{n_j}{n_j+1} \right), \quad a_i^* = \prod_{j=1}^{i} \left(\frac{n_j+1}{n_j+2} \right), \quad x_{(i)} = 2z_{(i)} + s_{(i)} \quad (i = 1, \ldots, k).$$

The extreme minimum value distribution gives scores (23) and a variance estimator (Cox, 1972) of

$$V_0 = \sum_{i=1}^{k} \left\{ n_i^{-1} \sum_{l \in R(w_{(i)})} z_l z_l^{\mathrm{T}} - n_i^{-2} \left(\sum_{l \in R(w_{(i)})} z_l \right) \left(\sum_{l \in R(w_{(i)})} z_l^{\mathrm{T}} \right) \right\}. \tag{29}$$

3·4. *Approximate scores*

The formulae (17) are inconvenient for many score generating densities, f, so that approximate scores, of the type (8), may be preferred. A first-order approximation to $\phi(u)$ about $u_{(i0)} = 1 - \hat{F}_W(w_{(i)})$ gives

$$J\{\phi(u_i)\} \simeq \phi(u_{i0}) + J\{(u_i - u_{i0})\}\,\phi'(u_{i0}) = \phi(u_{i0}).$$

A similar approximation to $\Phi(u)$ suggests the scoring scheme

$$c_i = \phi\{1 - \hat{F}_W(w_{(i)})\}, \quad C_i = \Phi\{1 - \hat{F}_W(w_{(i)})\}. \tag{30}$$

A first-order expansion of $\phi(u_i)$ gives

$$\begin{aligned} [J\{\phi(u_i)\} - \phi(u_{i0})]^2 &= [J\{(u_i - u_{i0})\,\phi'(u_i^*)\}]^2 \\ &\leqslant J\{(u_i - u_{i0})^2\}\,J\{\phi'(u_i^*)^2\} \\ &= \left\{ \prod_{j=1}^{i} \left(\frac{n_j+1}{n_j+2} \right) - \prod_{j=1}^{i} \left(\frac{n_j}{n_j+1} \right) \right\} \hat{F}_W(w_{(i)})\,J\{\phi'(u_i^*)^2\}, \end{aligned} \tag{31}$$

where u_i^* is between u_i and u_{i0}. A similar expression holds for the difference between censored data scores. Because of the first term on the right-hand side of (31), a sequence of $w_{(i)}$ values with fixed finite limit w, as sample size increases, are readily seen to give rise to convergent scores (30) and (17) under mild restrictions on the censoring. The more important problem of asymptotic equivalence of the corresponding test statistics is more complex and will not be considered here.

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As an example of the scoring system (30), a double exponential density leads to the following censored data generalization of the sign test:

$$c_i = \begin{cases} -1 & \text{if} \quad \hat{F}_W(w_{(i)}) > \frac{1}{2}, \\ 0 & \text{if} \quad \hat{F}_W(w_{(i)}) = \frac{1}{2}, \\ 1 & \text{if} \quad \hat{F}_W(w_{(i)}) < \frac{1}{2}; \end{cases} \quad C_i = \begin{cases} \dfrac{1-\hat{F}_W(w_{(i)})}{\hat{F}_W(w_{(i)})} & \text{if} \quad \hat{F}_W(w_{(i)}) > \frac{1}{2}, \\ 1 & \text{if} \quad \hat{F}_W(w_{(i)}) \leqslant \frac{1}{2}. \end{cases}$$

3·5. *Asymptotic properties*

No formal discussion of asymptotic properties of (13) will be undertaken here. It is clear, however, that conditions on the regression vector z for asymptotic normality of (13) and for consistency of V_0 in (25) will be similar to those of Hájek & Šidák (1967, p. 152). Essentially it will be required that each observation have an asymptotically trivial relative contribution to (13). Asymptotic approximations may then be suspect if the sample contains extreme and isolated z values that individually can have a dominating effect of (13) or (25). Similarly, conditions for asymptotic normality and consistency would prevent censoring times from converging too rapidly to zero as this may leave a few observations with an asymptotically nonnegligible effect on (13) or (25).

Assuming such conditions are satisfied, the statistic

$$v^{\mathrm{T}} V_0^{-1} v \tag{32}$$

will have an asymptotic χ^2 distribution, under $\beta = \beta_0$, with degrees of freedom the dimension of z, provided that V_0 is nonsingular.

In comparison with the scores (17) or (30) the parametric scores (4) can be written

$$c_i = \phi\{1-F_W(w_{(i)})\}, \quad C_{ij} = \Phi\{1-F_W(w_{(ij)})\}.$$

Efficiency properties of the statistics corresponding to (17) and (30), under the assumed F, will then depend directly on the properties of the corresponding estimators of F. J. Crowley and D. R. Thomas have in a University of Wisconsin technical report studied the log rank test (23) under a random-censorship K-sample problem and found that some loss of asymptotic efficiency was experienced if the censoring distribution differed among the samples. This loss is presumably due to the inability of the present methods to utilize the ordering of censored observations that occur between adjacent uncensored values. One might, however, expect such loss to be slight unless the censoring depends on z in an extreme manner.

4. Discussion

4·1. *Data transformations*

An important aspect of rank tests is their invariance under monotone increasing transformations on the response variable. For example, if $w = y - \beta_0 z$ in (1) were replaced by some $w' = h(w)$, with h monotone increasing, the rank vector (10) and scores (17) would be unchanged. This feature is most easily appreciated in terms of a test of $\beta = 0$. Monotone transformations on y itself, then, do not affect the test procedure so that equivalent tests arise from taking y to be normal or log normal, for example. Consider a further example in the survival analysis setting. Cox (1972) introduced the proportional hazards family with hazard function at time t and regression vector z of the form

$$\lambda_0(t)\, e^{\beta z}. \tag{33}$$

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The test for $\beta = 0$, at any $\lambda_0(t)$, can be seen to lead to extreme value error in (1) and to the log rank statistic (23) simply by defining y, in (1), by

$$y = \log \int_0^t \lambda_0(u)\, du.$$

Such transformations are related to the conversion function of Peto & Peto (1972). Peto & Peto's methods may be used to generate tests in situations that cannot be reduced to (1) by suitable transformations.

4·2. *Survival data methods*

Survival studies provide one of the most frequent sources of arbitrary censored data. A second interesting partially parametric class of models is given by a hazard function

$$\tilde{\lambda}_0(t\, e^{\beta z})\, e^{\beta z}. \tag{34}$$

This time the regressor variable is assumed to affect failure time itself multiplicatively so that the model may be termed an accelerated failure time model. In contrast (32) assumes that z affects the hazard function multiplicatively. The two models intersect in the two-parameter Weibull model having

$$\lambda_0(t) = QP(Qt)^{P-1}$$

for Q and $P > 0$. A transformation to $y = \log t$ in (34) gives the model (1) with arbitrariness in $\tilde{\lambda}_0$ corresponding to that in (α, σ) and f.

With models (34) close to the Weibull form, the log rank test (23) is expected to have better efficiency properties than the corresponding test for $\beta = \beta_0 \neq 0$ from the proportional hazards model. More explicitly, the proportional hazards test is based on the generalized rank vector (10) of $y = \log t$ (Kalbfleisch & Prentice, 1973), rather than of $w = y - \beta_0 z$. The generalized rank vector for y becomes more predictable, and less informative, as β_0 departs from zero. Consider, for example, the case of an exponential regression model, $\lambda_0(.) = \tilde{\lambda}_0(.) = Q$. In this special case the meaning of β agrees in (33) and (34). With no censoring and scalar regression variable, the log rank test (24) based on $w - \beta_0 z$ is fully efficient from (9), while the corresponding proportional hazards test has efficiency of the form $\exp(-\beta_0^2 \mu_2)$, where μ_2 is the second moment of the finite population of z values (Kalbfleisch, 1974).

4·3. *Estimation*

The approximate χ^2 statistic (32) can also form the basis for more general inference on β. For example, an approximate confidence region for β is given by those β_0 values for which (32) does not exceed specified percentage points of the χ^2 distribution. An estimator $\tilde{\beta}$ may be defined as the β_0 value, or values, that minimize (32). Jurečková (1969, 1971) has shown that, with uncensored data, this estimator $\tilde{\beta}$ is fully efficient under the assumed score generating density and that its efficiency, more generally, is the same as that of the corresponding test, as given in (9).

Partial tests on β are frequently important. Suppose $\beta z = \beta_1 z_1 + \beta_2 z_2$ and it is desired to test $\beta_1 = \beta_1^0$. Such a test may be based on (13) as applied to residuals $w = y - \beta_1^0 z_1 - \tilde{\beta}_2^0 z_2$, where $\tilde{\beta}_2^0$ minimizes (32) subject to $\beta_1 = \beta_1^0$. Unfortunately, numerical aspects have not been adequately addressed in the computation of $\tilde{\beta}_2^0$ though a straightforward search may be used for scalar z_2.

An alternative that will sometimes be adequate is to stratify the sample on the basis of values of the nuisance regressor variable, z_2, and to apply the test (13) within each stratum. Let v_j represent the test statistic (13) and V_{0_j} the corresponding variance estimator (25), within stratum j. Assuming adequate homogeneity with respect to z_2 within strata, an overall

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test for $\beta_1 = \beta_1^0$ can be based on an approximate χ^2 distribution for

$$\Sigma v_j^{\mathrm{T}}(\Sigma V_{0_j})^{-1} \Sigma v_j, \tag{35}$$

where the summation is over strata and the degrees of freedom, assuming ΣV_{0_j} nonsingular, is the dimension of z_1. In the special case of log rank statistics v_j, (35) is sometimes referred to as the Mantel–Haenszel test after Mantel & Haenszel (1959), Mantel (1963, 1966).

4·4. *Matched pairs*

A rather extreme application of (35) is to matched pair data in which each pair defines a stratum. In pairs in which a censored w value precedes any uncensored values, $v_j = 0$ and $V_{0j} = 0$, a zero matrix. Otherwise, the rank vector is completely determined. Model (1) can be written

$$y_{ij} = \alpha + \beta z_{ij} + \sigma e_{ij}$$

for individual i ($i = 1, 2$), within stratum j, where e_{ij} has density f. Let $w_{ij} = y_{ij} - \beta_0 z_{ij}$, $d_j = z_{1j} - z_{2j}$ and

$$H_j = \begin{cases} -1 & (w_{2j} < w_{1j}), \\ 1 & (w_{1j} < w_{2j}). \end{cases}$$

Conditional on the smallest value of w in the pair being uncensored the statistic v_j is $v_j = d_j H_j c_f$, where c_f, or c_1 in above notation, is a constant depending on f. A permutation variance estimator for v_j is $d_j d_j^{\mathrm{T}} c_f^2$, so that (37) becomes

$$\Sigma_j (d_j^{\mathrm{T}} H_j)(\Sigma_j d_j d_j^{\mathrm{T}})^{-1} \Sigma_j (d_j H_j)$$

independently of the choice of generating distribution f. The asymptotic normality of $\Sigma d_j H_j$ follows under mild restrictions on d_j from the fact that H_j takes on its two values each with probability $\frac{1}{2}$ under $\beta = \beta_0$, even when restriction is made to pairs in which the smallest value of w is uncensored.

4·5. *Tied data*

The above methods assume throughout that uncensored values of w are distinct. Tied uncensored values at the same z may be broken arbitrarily because their ordering does not affect the test statistic (13). More generally, if the fraction of tied values of w is small, tied individuals may be assigned the average of their scores with a similar averaging in the variance matrix. This follows from redefining the rank vector (10) as consisting of all possible underlying rank vectors consistent with the observed tied and censored data (Peto, 1972b; Kalbfleisch & Prentice, 1973). The same idea allows the methods of this paper to be generalized, at least in principle, to doubly censored data.

5. Illustration

A rather simple illustration is given by the data of Table 1 (Pike, 1966). Times from insult with a carcinogen to the development of vaginal cancer are given for two groups of female rats. The groups were distinguished by a pretreatment regimen. In the above framework the response y may be taken to be the logarithm of time and we may set the single regressor variable to have value 0 for group 1 and value 1 for group 2. A test for equality of the two times to incidence distributions then involves testing $\beta = 0$. The generalized rank vector $r(w)$ is that based on log times or, because the transformation is monotone, times to incidence themselves. The n_j are simply the number of animals alive just before the jth smallest failure occurs, so that $n_1 = 40$, $n_2 = 39, \ldots$. Since tied times occur only at the same z value

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they may be ordered arbitrarily. The generalized Wilcoxon scores (23) give a test statistic $v = 2{\cdot}887$ and a variance estimator $V_0 = 3{\cdot}066$, so that the $\chi^2_{(1)}$ statistic (32) has value 2·72. In comparison the log rank scores give $v = 4{\cdot}584$, $V_0 = 7{\cdot}653$ and a $\chi^2_{(1)}$ statistic (32) of nearly identical value 2·75.

Table 1. *Days to vaginal cancer in female rats*

Group 1	143, 164, 188, 188, 190, 192, 206, 209, 213, 216, 220, 227, 230, 234, 246, 265, 304, 216+, 244+
Group 2	142, 156, 163, 198, 205, 232, 232, 233, 233, 233, 233, 239, 240, 261, 280, 280, 296, 296, 323, 204+, 344+

+: Censored.

The absence of small incidence times in Table 1 suggests the possible inclusion of a guarantee time in the model (1) to obtain a better fit. For instance, we may take $y = \log(t-\Delta)$ rather than $\log t$ as above. Again, since $g(y) = \log(e^y - \Delta)$ is monotone increasing, the test statistics would not change under such remodelling. Substantial changes could result, however, from such a change in a parametric analysis. Further, with specified Δ, a parametric analysis may be quite sensitive to the choice of f. For example, maximum likelihood estimation as applied to a Weibull regression model for $t-100$; or equivalently (1) with extreme minimum value error for $y = \log(t-100)$ gives $\hat{\beta} = 0{\cdot}213$ with an estimated standard error of 0·01162. A $\chi^2_{(1)}$ statistic for testing $\beta = 0$ then has value $(0{\cdot}213)^2/(0{\cdot}01162) = 3{\cdot}90$. By contrast, a log normal model for $(t-100)$, or a normal error for $y = \log(t-100)$, gives $\hat{\beta} = 0{\cdot}154$ with estimated standard error 0·01798, so that $\chi^2_{(1)} = (0{\cdot}154)^2/(0{\cdot}01795) = 1{\cdot}32$. Parametric error assumptions thus give rise to very different inferences on β. Note that if the Weibull model can be substantiated it would give rise, in this case, to somewhat more significant results than do the rank tests mentioned previously.

Finally, still with $y = \log(t-100)$ the $\chi^2_{(1)}$ statistic (32) was scanned over β values. An approximate 95% confidence interval for β is $(-0{\cdot}025, 0{\cdot}414)$ using generalized Wilcoxon scores (22) and $(-0{\cdot}038, 0{\cdot}424)$ using log rank scores (23). The corresponding estimators $\tilde{\beta}$ at which (13) changes sign or (32) is minimized each have value 0·180, intermediate to those from the parametric analyses.

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REFERENCES

ALTSHULER, B. (1970). Theory for the measurement of competing risks in animal experiments. *Math. Biosci.* **6**, 1–11.

BIRNBAUM, A. & LASKA, E. (1967). Efficiency robust two-sample rank tests. *J. Am. Statist. Assoc.* **62**, 1241–51.

BRESLOW, N. (1970). A generalized Kruskal–Wallis test for comparing k samples subject to unequal patterns of censorship. *Biometrika* **57**, 579–94.

COX, D. R. (1972). Regression models and life tables (with discussion). *J. R. Statist. Soc.* B **34**, 187–220.

COX, D. R. & HINKLEY, D. V. (1974). *Theoretical Statistics*. London: Chapman & Hall.

CROWLEY, J. (1974). A note on some recent likelihoods leading to the log rank test. *Biometrika* **61**, 533–8.

FISHER, R. A. & YATES, F. (1963). *Statistical Tables for Biological, Agricultural and Medical Research*, 6th edition. Edinburgh: Oliver & Boyd.

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GASTWIRTH, J. L. (1970). On robust rank tests. In *Nonparametric Techniques in Statistical Inference*. Ed. M. L. Puri, pp. 89–109. Cambridge University Press.

GEHAN, E. (1965). A generalized Wilcoxon test for comparing arbitrarily singly censored samples. *Biometrika* **52**, 203–23.

HÁJEK, J. & ŠIDÁK, Z. (1967). *Theory of Rank Tests*. New York: Academic Press.

JOHNSON, R. A. & MEHROTRA, K. G. (1972). Locally most powerful rank test for the two-sample problem with censored data. *Ann. Math. Statist.* **43**, 823–31.

JUREČKOVÁ, J. (1969). Asymptotic linearity of a rank statistic in regression parameter. *Ann. Math. Statist.* **40**, 1889–900.

JUREČKOVÁ, J. (1971). Nonparametric estimate of regression coefficients. *Ann. Math. Statist.* **42**, 1328–38.

KALBFLEISCH, J. D. (1974). Some efficiency computations for survival distributions. *Biometrika* **61**, 31–8.

KALBFLEISCH, J. D. & PRENTICE, R. L. (1973). Marginal likelihoods based on Cox's regression and life model. *Biometrika* **60**, 267–78.

KAPLAN, E. L. and MEIER, P. (1958). Nonparametric estimation from incomplete observations. *J. Am. Statist. Assoc.* **53**, 457–81.

MANTEL, N. (1963). Chi-square tests with one degree of freedom: extensions of the Mantel–Haenszel procedure. *J. Am. Statist. Assoc.* **58**, 690–700.

MANTEL, N. (1966). Evaluation of survival data and two new rank order statistics arising in its consideration. *Cancer Chemotherapy Rep.* **50**, 163–70.

MANTEL, N. & HAENSZEL, W. (1959). Statistical aspects of the analysis of data from retrospective studies of disease. *J. Nat. Cancer Inst.* **22**, 719–48.

PETO, R. (1972a). Rank tests of maximal power against Lehmann-type alternatives. *Biometrika* **59**, 472–5.

PETO, R. (1972b). Contribution to discussion of paper by D. R. Cox. *J. R. Statist. Soc.* B **34**, 205–7.

PETO, R. & PETO, J. (1972). Asymptotically efficient rank invariant test procedures (with discussion). *J. R. Statist. Soc.* A **135**, 185–206.

PIKE, M. (1966). A method of analysis of a certain class of experiments in carcinogenesis. *Biometrics* **22**, 142–61.

PURI, M. L. & SEN, P. K. (1971). *Nonparametric Methods in Multivariate Analysis*. New York: Wiley.

SAVAGE, I. R. (1956). Contributions to the theory of rank order statistics—the two-sample case. *Ann. Math. Statist.* **27**, 590–615.

VAN DER WAERDEN, B. L. (1953). Ein neuer Test für das Problem der zwei Stichproben. *Math. Annalen* **126**, 93–107.

WILCOXON, F. (1945). Individual comparisons by ranking methods. *Biometrics* **1**, 80–3.

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